LING285-L1

Course: LING285: Human Language and Technology
Title of Lecture: Lecture 1 - Human Language and Technology
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Section 1: Watching the Vocal Tract During Speech

- **Real-time magnetic resonance imaging (real-time MRI)** records a moving picture of the inside of a person's head and neck while they talk. The view is taken from the side, as if the head were cut straight down the middle from the forehead to the chin, so the face points to the left and the back of the neck is on the right. Air shows up black. Soft tissue such as the tongue, lips, and throat walls shows up in shades of gray and white. This method lets us watch structures move that are normally hidden inside the body.
    - The **lips** sit at the far left edge, and they open, close, round, and spread during speech.
    - The **tongue** is the large rounded gray mass that fills most of the mouth. It is the most active structure in speech, and it changes shape constantly, bunching up toward the roof of the mouth, pulling back toward the throat, or lying low.
    - The **roof of the mouth** runs above the tongue. Its front part is the hard, bony **hard palate**. Its back part is the soft, movable **soft palate**, also called the **velum**, which can lift to seal off the nose or lower to let air flow out through the nose.
    - Behind the tongue, a dark vertical channel of air runs down the throat. This is the **pharynx**. Below it sits the **larynx**, which holds the **vocal folds**, and behind all of it the bright vertical column of the spine runs down the right side of the image.
- When a person speaks, all of these structures move at the same time and at high speed. The jaw rises and falls, the lips open and close, the tongue reshapes itself many times per second, and the soft palate lifts and lowers. Each sound we make depends on a precise shape of this whole passage, and the shapes follow one another smoothly rather than one at a time. The two MRI movies capture the same speaker's vocal tract in two different shapes, and the difference between them comes almost entirely from where the tongue and lips are placed.

Section 2: How Speech and Reading Are Learned

- We learn to move our speech organs in extremely complex ways with no instructions and with almost no visual feedback. Nobody tells a young child where to place the tongue tip for a particular sound or how high to lift the back of the tongue. A child also cannot see their own tongue, soft palate, or vocal folds, since these are hidden inside the mouth and throat in exactly the way the MRI images reveal. Even so, children master these movements.
- **By ages 2 to 4**, children speak with a wide vocabulary and with language that is close to adult language. This happens with hardly any explicit instruction. Children are not given lessons in grammar or pronunciation at this age, yet they pick up thousands of words and the rules for putting them together just from being around people who talk.
- **Reading and writing** follow a very different path. A child may memorize the visual shapes of the letters of the alphabet fairly early. Reading and writing whole words, though, usually comes at **ages 6 to 7**, and it requires a great deal of instruction.
    - The contrast is striking. Speech involves hidden organs that the learner cannot see, yet it develops almost on its own.
    - Reading and writing involve the eyes and hands, which are fully visible, yet they need years of deliberate teaching.
    - This difference shows that speech is a deep human cognitive skill that the brain is prepared to acquire, while literacy is a learned tool built on top of it.

Section 3: Articulatory Phonetics and Speech Technology

- **Articulatory phonetics** is the field that studies the movements of the different structures of the vocal tract. The **vocal tract** is the full air passage used in speech, running from the larynx up through the throat and mouth to the lips, with a side branch into the nose. Articulatory phonetics asks which structures move, where they move to, and how those movements create the different sounds of a language. Real-time MRI is one of the main tools of this field, since it lets researchers measure the tongue, lips, and soft palate directly while a person talks.
- In just the **last two years**, **voice assistants** and other **speech technology** have come to **emulate** human speech ability in ways that almost no one thought possible. To emulate an ability means to reproduce its behavior. These systems can now produce speech that sounds natural and humanlike, and they can recognize and respond to spoken language with high accuracy.
- **LING 285 is about both human speech and AI speech.** One side of the course studies how people physically produce and perceive speech. The other side studies how modern AI systems model that same ability, so the human skill and the machine version can be compared directly.

Section 4: The Vocal Tract as a Vibrating Tube of Air

- The next step in understanding speech is to focus not on the tongue and lips themselves but on the **air** that fills the space between the tongue and the rest of the vocal tract. In the real vocal tract, that air space bends: it runs down the throat and then turns forward through the mouth. If we ignore that curve and straighten the passage out, the vocal tract becomes a simple **tube of air**.
    - One end of the tube corresponds to the **lips**, where the passage is open to the outside air.
    - The other end corresponds to the **larynx** deep in the throat, where the tube is treated as closed.
    - The walls of the tube stand for the tongue and the other surfaces that surround the air space.
- The tube is labeled **schwa**. Schwa is the neutral, relaxed vowel sound, the "uh" sound in the first syllable of "about" or the last syllable of "sofa." For schwa, the tongue sits in a central resting position and does not narrow the passage much anywhere, so the air space is close to the same width from end to end. That makes schwa the vowel that fits the straight, even tube best.
- Inside the tube, the air is drawn as many small blue circles arranged in vertical columns, and each circle stands for a small bit of air. When the tube vibrates, these bits of air do not travel down the tube and out the end. Instead, each column shifts back and forth around its own resting position.
    - As neighboring columns move toward each other, the air in that region is squeezed together into a **compression**, where the air is packed more tightly.
    - As neighboring columns move away from each other, the air in that region spreads out into a **rarefaction**, where the air is packed more loosely.
    - These regions of packing and spreading move along the tube and reflect back from its ends, and this repeating pattern is the **vibration of air** that makes up the sound of the vowel.
- **Acoustic phonetics** is the field that studies exactly this: how air vibrates in the vocal tract and what sound those vibrations produce. Articulatory phonetics describes the shape of the tube, and acoustic phonetics describes the physics of the air inside it. The two fields connect directly, since changing the tongue and lip positions changes the shape of the tube, and changing the shape of the tube changes how the air inside it vibrates.

Section 5: Harms, Benefits, and the Problem of Understanding AI

- AI brings real **psychological, economic, social, and environmental harms**. Given everything known about these harms, it is tempting to simply ignore AI and act as if it did not exist. Many scientists also warn about **existential harm**, meaning the possibility that advanced AI could threaten humanity's survival or basic way of life. Even with that warning, pretending AI does not exist is not a workable option.
- AI exists, and it comes with many harms and also some benefits. Two examples of benefits are **immunotherapy**, a medical approach that uses the body's own immune

system to fight diseases such as cancer, and **environmental engineering**, the design of solutions to environmental problems.
    - The environmental case cuts both ways, as captured by a newspaper headline asking, **"Will A.I. Ruin the Planet or Save the Planet?"**
    - Running large AI systems uses enormous amounts of energy, which is one of the environmental harms, while AI tools are also being applied to engineering problems that could help the environment.
- One of the biggest dangers of AI is that **we have no idea how it works**. This is summed up in the phrase **"We know everything, and understand nothing."**
    - "We know everything" means that every number and every calculation inside an AI system is available to be inspected, since people built the system and can look at all of its parts.
    - "We understand nothing" means that, even with all of those details in hand, no one can explain why the system produces the particular answers it does or what it is really doing when it produces them.
- The number of people who **know** how these systems operate is very, very small. That concentration of knowledge is a danger in itself, since the decisions about how AI is built and used end up in very few hands.
- **AI literacy**, meaning a working understanding of what AI systems are and how they behave, is necessary for many groups of people. Scientists who use AI need it to judge whether their results are trustworthy. Politicians who need to control AI need it to write sensible rules. Lawyers who litigate cases involving AI need it to argue about what a system did and why.
- This course is built on the belief that a **creative thinker** in a class like this one can help us understand these systems well enough to benefit from them while avoiding their negative consequences. Understanding is the path to both getting the benefits and limiting the harms.

Section 6: Goals of the Course

- The course has four **specific goals**.
    - The first is to appreciate the human cognitive skill of **speech**, the complex ability described in the MRI images and the learning timeline, which people acquire with almost no instruction.
    - The second is to understand some basic principles of **physics** through speech, such as the way air vibrates inside the tube of the vocal tract.
    - The third is to understand how modern **AI systems model human cognition**, meaning how machines reproduce abilities such as speaking and understanding language.
    - The fourth is to sharpen **methodical reasoning** through **elementary programming**. Writing simple programs forces each step of a problem to be stated clearly and in the correct order, which trains careful, step-by-step thinking.
- The course also has three **general goals**.
    - The first is learning to think creatively in a rich **interdisciplinary** way. The material draws on physics, psychology, linguistics, biology, economics, computer science, philosophy, neuroscience, and many other fields. Speech alone already touches several of these, since it involves the biology of the vocal tract, the physics of vibrating air, the psychology of learning, and the neuroscience of how the brain controls movement.
    - The second is deepening our understanding of the world we are all living in today, a world in which AI systems are becoming part of daily life.
    - The third is sharpening our **skillful use of modern AI systems**, so that we use them well and with a clear sense of what they can and cannot do.

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Section 7: Diotima's Ladder and What Is Valued in 2026

- **Diotima**, dated here to around **440 BCE**, is the teacher who, in Plato's dialogue the *Symposium*, teaches **Socrates** about a ladder of ascent. The ladder describes how a

person climbs from ordinary, surface-level knowledge of particular things toward deeper understanding of the general ideas behind them. A chain of teachers runs from her through Greek philosophy: Diotima taught Socrates, Socrates taught **Plato**, and Plato taught **Aristotle**. The large marble bust represents Diotima, and the three smaller portraits below her represent Socrates, Plato, and Aristotle in that order.

- The **bottom of the ladder** holds the kinds of knowledge and skill that come from routine and from copying what already exists.
  - These include **daily experience**, **habit**, and **memorization**, along with **specialized skills**, **repetition and replication**, and learning from **textbooks**.
  - Thinking at this level follows the rule that **similar is same, and slight difference is different**. Things that look alike on the surface are treated as identical, and things that look even a little different on the surface are treated as unrelated.
- The **top of the ladder** holds **creativity, insight, thought, focus, attention, imagination, and vision**, along with **problem-posing, problem-solving, critical reading, and creative writing**.
  - Thinking at this level follows a reversed rule: **different can be same, and similar can be different**. Things that look different on the surface can share the same deeper structure, and things that look alike on the surface can turn out to be fundamentally different.
  - Speech gives a direct example. The sounds f and v sound different, yet they are made in the same place in the mouth, so in that respect they are the same. The th in "thing" and the th in "that" are spelled identically, yet one uses vocal fold vibration and the other does not, so they are different sounds.
  - **Problem-posing** means asking good questions in the first place, which is separate from **problem-solving**, which means answering questions once they have been asked.
- A person climbs the ladder through repeated cycles of **question and answer**. Each cycle of asking a question and working out an answer lifts understanding one rung higher, and the cycles repeat again and again on the way from the bottom to the top. This is the method Socrates became famous for, and it traces back to Diotima's teaching of him.
- For about **2500 years**, people could fake it. They could do well in school and at work without ever climbing the ladder, simply by memorizing, repeating, and replicating. In **2026** that is no longer true. In interviews for jobs and for graduate school, what matters now is whether a person shows the properties at the **top of Diotima's ladder**, such as creativity, insight, problem-posing, and the ability to see when different things are the same and similar things are different.

Section 8: Learning Speech by Doing: Voice and Place of Articulation

- The quickest way to understand speech sounds is to produce them and pay attention to what the body does. The exercise uses four pairs of sounds, and each pair is held out as one long continuous sound that switches partway through from the first sound to the second.
  - The first pair is **f then v**. Say the f in "fat" and hold it, then switch to the v in "vat" and hold that.
  - The second pair is **th then th**. Say the th in "thing" and hold it, then switch to the th in "that" and hold that.
  - The third pair is **s then z**. Say the s in "sing" and hold it, then switch to the z in "zing" and hold that.
  - The fourth pair is **sh then j**. Say the sh in "mesh" and hold it, then switch to the sound written j here, which is the sound in the middle of "beige." This is the soft buzzing sound, not the j in "jump."
- The first question is what changes between the first and second sound of each pair. Resting the fingers lightly on the front of the throat while switching makes the change easy to feel.
  - During f, the first th, s, and sh, the throat stays still and the sound is a pure hiss of air.
  - During v, the second th, z, and the sound in "beige," a buzzing vibration appears in the throat and the sound gains a humming quality.

    - The difference is **voice**, which is whether the **vocal folds** vibrate or not. The vocal folds are two small folds of tissue inside the larynx. When they are held apart, air passes through silently and the sound is **voiceless**. When they are brought close together, the air flowing between them makes them vibrate rapidly, and the sound is **voiced**.
    - Within each pair, the tongue and lips stay in the same position the whole time. Only the vocal folds change, which shows that voicing is controlled separately from the mouth.
- The second question is what changes from one line to the next, meaning from the f and v pair to the th pair to the s and z pair to the sh pair.
    - For f and v, the lower lip presses lightly against the upper front teeth.
    - For both th sounds, the tip of the tongue touches the upper front teeth.
    - For s and z, the tip of the tongue comes close to the ridge just behind the upper front teeth.
    - For sh and the sound in "beige," the tongue pulls a bit further back, behind that ridge.
    - The difference is **place of articulation**, which is the location in the vocal tract where the narrowing that makes the sound is formed. Moving from one line to the next moves that narrowing steadily further back in the mouth.
- Voice and place of articulation are two separate, independent properties of a speech sound. Every sound in this exercise can be described by stating both of them, and changing one while keeping the other fixed produces a different sound.

Section 9: Fricatives

- All of the sounds in the exercise belong to a group called **fricatives**. A fricative is made by bringing two parts of the vocal tract very close together without closing the passage completely. Air pushed from the lungs is forced through the narrow gap, and passing through such a small opening makes the air flow turbulent and chaotic. That turbulent flow produces the hissing or buzzing noise that is typical of fricatives, and the sound can be held for as long as there is breath.
- The English fricatives are organized in a table by **voice** in the rows and **place of articulation** in the columns.

	Labial	**Dental**	**Alveolar**	**Postalveolar**
Voiceless	f	th	s	sh
Voiced	v	th	z	j

- Each column is a place of articulation, and each column contains one voiceless and one voiced fricative made in that same place.
    - **Labial** means made with the lips. For f and v, the lower lip rises to lightly touch the upper front teeth, and the air hisses through the gaps between them. Real-time MRI of these sounds shows the lower lip tucked up under the upper teeth while the tongue lies relatively low and relaxed in the mouth.
    - **Dental** means made at the teeth. For the th in "thing" and the th in "that," the tip of the tongue touches or rests just between the upper front teeth. The MRI of these sounds shows the front of the tongue pushed forward right up to the teeth.
    - **Alveolar** means made at the **alveolar ridge**, the small bony ridge on the roof of the mouth just behind the upper front teeth. For s and z, the tongue tip rises to form a narrow groove close to that ridge. The MRI shows the tongue tip raised toward the front of the roof of the mouth, a short distance behind the teeth.
    - **Postalveolar** means made just behind the alveolar ridge. For sh and the sound in "beige," a wider section of the front of the tongue rises toward the roof of the mouth a little further back than for s. The MRI shows the raised part of the tongue shifted backward compared with s, and the lips often push slightly forward during these sounds.

- Each row is a voicing category. The top row holds the **voiceless** fricatives, made with the vocal folds apart and still. The bottom row holds the **voiced** fricatives, made with the vocal folds vibrating. Two fricatives in the same column differ only in voicing, and two fricatives in the same row differ only in place of articulation. The two th sounds make the point sharply, since English spells them the same way even though one is voiceless and the other is voiced.

Section 10: Stops

- **Stops** are made differently from fricatives. For a stop, two parts of the vocal tract come together and **close the passage completely**, so that for a brief moment air cannot escape through the mouth at all.
    - While the closure is held, air coming up from the lungs keeps pushing into the mouth, and the air pressure builds up behind the closure.
    - When the closure is released, the trapped air bursts out suddenly, and that burst is part of what makes a stop sound short and sharp.
    - Unlike a fricative, a stop cannot be held as a continuous sound, since the sound depends on the closing and the release.
- The English stops fall into a grid of three places of articulation and three types. Real-time MRI movies capture each of these nine sounds in the middle of its closure.
- The first row is **p, b, and m**, and all three are made by closing the **lips**.
    - For p, the lips press together, the vocal folds stay still, and the release produces a voiceless puff of air.
    - For b, the lips close in the same way, but the vocal folds vibrate, so b is voiced.
    - For m, the lips close in the same way and the vocal folds vibrate, but the **soft palate lowers**. This opens the passage into the nose, so air flows out through the nose the whole time the lips are closed. A sound made this way is called a **nasal**, and the MRI of m shows the gap opened between the soft palate and the back of the throat.
- The second row is **t, d, and n**, and all three are made by pressing the tip of the tongue against the **alveolar ridge**, the same place used for s and z.
    - For t, the tongue tip seals against the ridge and the vocal folds stay still, so t is voiceless.
    - For d, the tongue tip seals in the same place and the vocal folds vibrate, so d is voiced.
    - For n, the tongue tip seals in the same place, the vocal folds vibrate, and the soft palate lowers so air escapes through the nose, which makes n a nasal. The MRI images of this row show the front of the tongue raised and pressed against the roof of the mouth just behind the teeth.
- The third row is **k, g, and ng**, and all three are made by raising the **back of the tongue** against the **soft palate** at the back of the roof of the mouth.
    - For k, the back of the tongue seals against the soft palate and the vocal folds stay still, so k is voiceless.
    - For g, the seal is in the same place and the vocal folds vibrate, so g is voiced.
    - For ng, the sound at the end of "sing," the seal is in the same place, the vocal folds vibrate, and the passage to the nose opens so air flows out through the nose. The MRI images of this row show the tongue bunched upward and backward, meeting the roof of the mouth far back near the throat, while the lips and tongue tip stay out of the way.
- The columns of the grid follow the same logic as the fricative table. The first column, **p, t, and k**, holds the **voiceless** stops. The second column, **b, d, and g**, holds the **voiced** stops. The third column, **m, n, and ng**, holds the **nasal** stops, which are voiced and send air through the nose. Reading down a column changes only the place of articulation, from the lips to the alveolar ridge to the soft palate. Reading across a row keeps the place fixed and changes only voicing and whether the nose is open. Nine different sounds come from combining just three places of closure with three settings of the vocal folds and soft palate, which shows how a small number of independent movements builds a large set of speech sounds.

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Section 1: Watching the Vocal Tract During Speech

- **Real-time magnetic resonance imaging (real-time MRI)** records a moving picture of the inside of the head and neck while a person talks. The view is a side view cut straight down the middle of the head, with the face pointing left and the back of the neck on the right. Air appears black, and soft tissue such as the tongue, lips, and throat walls appears gray and white.
  - The **lips** sit at the far left and open, close, and change shape during speech.
  - The **tongue** is the large rounded mass filling most of the mouth, and it constantly reshapes itself, rising toward the roof of the mouth, pulling back toward the throat, or lying low.
  - The roof of the mouth has a bony front part, the **hard palate**, and a soft, movable back part, the **soft palate**, which can lift to seal off the nose or lower to let air flow out through the nose.
  - Behind the tongue, a dark channel of air runs down the throat, called the **pharynx**. Below it sits the **larynx**, which holds the **vocal folds**.
- During speech, these structures all move at once and at high speed. The jaw rises and falls, the lips open and close, the tongue changes shape many times per second, and the soft palate lifts and lowers. Each speech sound depends on a precise shape of this whole passage.

Section 2: How Speech and Reading Are Learned

- We learn to move our speech organs in extremely complex ways with **no instructions** and with **almost no visual feedback**. Nobody tells a child where to place the tongue for a sound, and a child cannot see their own tongue, soft palate, or vocal folds, which stay hidden inside the mouth and throat. Children master these movements anyway.
- **By ages 2 to 4**, children speak with a wide vocabulary and with near-adult language, and this happens with hardly any explicit instruction.
- **Reading and writing** develop very differently. A child may memorize the visual shapes of the alphabet early, but reading and writing whole words comes at **ages 6 to 7** and requires a great deal of instruction.
  - Speech uses hidden organs the learner cannot see, yet it develops with almost no teaching.
  - Reading and writing use the eyes and hands, which are fully visible, yet they depend on extensive teaching.

Section 3: Articulatory Phonetics and Speech Technology

- **Articulatory phonetics** is the field that studies the movements of the different structures of the **vocal tract**. The vocal tract is the air passage used in speech, running from the larynx up through the throat and mouth to the lips, with a branch into the nose. Articulatory phonetics describes which structures move and where they move to when each sound is made, and real-time MRI makes these movements directly measurable.
- In the **last two years**, **voice assistants** and other **speech technology** have come to **emulate** human speech ability in ways almost no one thought possible. To emulate an ability means to reproduce its behavior, so these systems now produce and respond to spoken language in a humanlike way.
- The subject of this material is both **human speech** and **AI speech**: how people produce speech and how AI systems reproduce that same ability.

Section 4: The Vocal Tract as a Vibrating Tube of Air

- The **air** between the tongue and the rest of the vocal tract can be treated as its own object. In the real vocal tract this air space is curved, running down the throat and then forward through the mouth. Ignoring the curve and straightening the passage turns the vocal tract into a simple **tube of air**.
    - One end of the tube corresponds to the **lips**, where the passage is open.
    - The other end corresponds to the **larynx**, where the tube is closed.
    - The walls of the tube stand for the tongue and the other surfaces surrounding the air.
- The tube represents the vowel **schwa**, the neutral "uh" sound in the first syllable of "about." For schwa, the tongue sits in a central resting position and does not narrow the passage anywhere in particular, so the air space has close to the same width from end to end.
- Inside the tube, the air is represented as columns of small particles, and when the air vibrates, the columns shift back and forth around their resting positions rather than flowing out of the tube.
    - Where neighboring columns move toward each other, the air is squeezed together into a region of higher packing called a **compression**.
    - Where neighboring columns move apart, the air spreads out into a region of lower packing called a **rarefaction**.
    - The repeating pattern of compressions and rarefactions along the tube is the **vibration of air** in the vocal tract.
- **Acoustic phonetics** is the field that studies how air vibrates in the vocal tract. Articulatory phonetics describes the shape of the tube, and acoustic phonetics describes how the air inside that shape vibrates, so a change in tongue or lip position changes the vibration of the air.

Section 5: Harms, Benefits, and the Problem of Understanding AI

- AI causes known **psychological, economic, social, and environmental harms**, and many scientists also warn about **existential harm**, meaning a threat to humanity's survival or basic way of life. Given these harms, it is tempting to ignore AI and pretend it does not exist.
- AI does exist, with many harms and also some benefits. Two benefits are **immunotherapy**, a medical approach that uses the body's own immune system to fight disease, and **environmental engineering**. The environmental question cuts both ways, as expressed in the headline **"Will A.I. Ruin the Planet or Save the Planet?"**
- One of the biggest dangers of AI is that **we have no idea how it works**, summed up in the phrase **"We know everything, and understand nothing."**
    - "We know everything" means that every part and every operation inside an AI system is available to be inspected.
    - "We understand nothing" means that even with all of those details available, no one can explain how the system actually produces what it does.
- The number of people who **know** how these systems operate is very, very small.
- **AI literacy**, a working understanding of what AI systems are and how they behave, is necessary for scientists who use AI, for politicians who need to control it, and for lawyers who litigate cases involving it. A creative thinker who understands these systems well enough can help society gain their benefits while avoiding their negative consequences.

Section 6: Diotima's Ladder and What Is Valued in 2026

- **Diotima**, dated to **440 BCE**, is associated with a **ladder** that describes the climb from surface-level knowledge to deep understanding. A line of Greek thinkers follows her: **Socrates**, **Plato**, and **Aristotle**.
- The **bottom of the ladder** holds **daily experience, habit, memorization, specialized skills, repetition and replication**, and **textbooks**. Thinking at this level follows the rule that **similar is same, and slight difference is different**, meaning things that look alike are treated as identical and things that look slightly different are treated as unrelated.

- The **top of the ladder** holds **creativity, insight, thought, focus, attention, imagination, vision, problem-posing, problem-solving, critical reading, and creative writing**. Thinking at this level follows the reversed rule that **different can be same, and similar can be different**.
    - Things that look different on the surface can share the same deeper structure. The sounds f and v sound different, but they are made in the same place in the mouth.
    - Things that look alike on the surface can differ fundamentally. The th in "thing" and the th in "that" are spelled the same, but one uses vocal fold vibration and the other does not.
    - **Problem-posing** means asking good questions, and **problem-solving** means answering them.
- A person climbs the ladder through repeated cycles of **question and answer**, with each cycle moving understanding one rung higher.
- For **2500 years**, people could fake it and do well without climbing the ladder. In **2026** that is no longer possible. In interviews for jobs and graduate school, what matters is whether a person shows the properties at the **top of Diotima's ladder**.

Section 7: Voice and Place of Articulation

- Producing sounds and paying attention to the body reveals how speech sounds differ. Each of four pairs is held as one long sound that switches partway through.
    - **f then v**: the f in "fat," then the v in "vat."
    - **th then th**: the th in "thing," then the th in "that."
    - **s then z**: the s in "sing," then the z in "zing."
    - **sh then j**: the sh in "mesh," then the sound written j, which is the sound in "beige," not the j in "jump."
- The first and second sounds of each pair differ in **voice**, which is whether the **vocal folds** vibrate or not.
    - For f, the th in "thing," s, and sh, the vocal folds do not vibrate, and these sounds are **voiceless**.
    - For v, the th in "that," z, and the sound in "beige," the vocal folds vibrate, and these sounds are **voiced**.
    - Within each pair, the mouth stays in the same position and only the vocal folds change, so voice is controlled separately from the mouth.
- Sounds on different lines differ in **place of articulation**, which is the location in the vocal tract where the narrowing that makes the sound is formed.
    - For f and v, the lower lip meets the upper front teeth.
    - For both th sounds, the tongue tip meets the upper front teeth.
    - For s and z, the tongue tip comes close to the ridge just behind the upper front teeth.
    - For sh and the sound in "beige," the tongue is further back, behind that ridge.
- Voice and place of articulation are independent properties, and every one of these sounds is described by stating both.

Section 8: Fricatives

- All of these sounds are **fricatives**. A fricative is made by bringing two parts of the vocal tract very close together without fully closing the passage, so air forced through the narrow gap produces a hissing or buzzing noise that can be held continuously.
- The fricatives are organized by voice in the rows and place of articulation in the columns.

	Labial	**Dental**	**Alveolar**	**Postalveolar**
Voiceless	f	th	s	sh
Voiced	v	th	z	j

- Each column is one place of articulation, and real-time MRI captures the vocal tract shape for each.

    - **Labial** means made with the lips. For f and v, the lower lip rises to the upper front teeth while the tongue stays relatively low.
    - **Dental** means made at the teeth. For both th sounds, the front of the tongue pushes forward to the upper front teeth.
    - **Alveolar** means made at the **alveolar ridge**, the bony ridge just behind the upper front teeth. For s and z, the tongue tip rises close to that ridge.
    - **Postalveolar** means made just behind the alveolar ridge. For sh and j, the raised part of the tongue sits further back than for s and z.
- Each row is one voicing category. Two fricatives in the same column differ only in voice, and two fricatives in the same row differ only in place of articulation.

Section 9: Stops

- **Stops** are made by closing the vocal tract passage completely, so air briefly cannot escape through the mouth. Air pressure builds up behind the closure, and the air bursts out when the closure is released.
- The stops form a grid of three places of closure and three types, and real-time MRI captures each one during its closure.
- **p, b, and m** are made by closing the **lips**.
    - For p, the vocal folds do not vibrate, so p is voiceless.
    - For b, the vocal folds vibrate, so b is voiced.
    - For m, the vocal folds vibrate and the soft palate lowers, so air flows out through the nose while the lips stay closed. A sound made with air flowing through the nose is a **nasal**.
- **t, d, and n** are made by pressing the tongue tip against the **alveolar ridge**, the same place used for s and z.
    - t is voiceless, d is voiced, and n is voiced and nasal, with the soft palate lowered so air leaves through the nose.
- **k, g, and ng** are made by raising the **back of the tongue** against the **soft palate**.
    - k is voiceless, g is voiced, and ng, the sound at the end of "sing," is voiced and nasal.
- The columns follow the same pattern as the fricatives. **p, t, and k** are voiceless, **b, d, and g** are voiced, and **m, n, and ng** are nasal. Moving down a column changes only the place of closure, from the lips to the alveolar ridge to the soft palate, and moving across a row keeps the place fixed while changing voicing and nasal airflow. Nine distinct stops come from combining three places of closure with three settings of the vocal folds and soft palate.